

Clarkson Institute Summer School: Examples Sheet 6 - Maps between groups

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Homomorphisms

1. (a) Find all the homomorphisms $f : C_6 \rightarrow C_2$ and $f : C_6 \rightarrow C_3$.
- (b) Find all the homomorphisms $f : D_3 \rightarrow C_2$ and $f : D_3 \rightarrow C_3$.

Solution:

- (a) For the first case $f : C_6 \rightarrow C_2$, let $C_6 = \{e, r, r^2, r^3, r^4, r^5\}$ and $C_2 = \{e, s\}$. Then $f(r)$ determines the homomorphism. If $f(r) = e$, then we have $f(r^k) = e$ for all k . If $f(r) = s$, we have $f(r^k) = s$ for odd k , $f(r^k) = e$ for even k .

In the case $f : C_6 \rightarrow C_3$, let $C_6 = \{e, r, r^2, r^3, r^4, r^5\}$ and $C_3 = \{e, s, s^2\}$. Once again, $f(r)$ determines the homomorphism. If $f(r) = e$, then we have $f(r^k) = e$ for all k . If $f(r) = s$, we have:

$$f(r^k) = f(r)^k = s^k.$$

Finally if $f(r) = s^2$, we have:

$$f(r^k) = f(r)^k = s^{2k}.$$

- (b) For a homomorphism $f : D_3 \rightarrow C_2$, we must say where the generators are mapped, $f(r)$, $f(m)$, and we must ensure that the relations are preserved, $f(r)^3 = e$, $f(m)^2 = e$ and $f(r)f(m) = f(m)f(r)^{-1}$. Since the image is Abelian, we note that the third relation can be rewritten as $f(r)^2 = e$ (by commuting $f(m)$ and cancelling, giving $f(r) = f(r)^{-1}$). The relations $f(r)^2 = e$ and $f(r)^3 = e$ simultaneously imply that $f(r) = e$. Thus the possible homomorphisms are generated by $f(m) = e$ and $f(m) = s$, given by:

$$f(r^a m^b) = e, \quad \text{or} \quad f(r^a m^b) = s^b.$$

On the other hand, for a homomorphism $f : D_3 \rightarrow C_3$, we require $f(r)^2 = e$ and $f(r)^3 = e$ simultaneously again, implying that $f(r) = e$. The requirement that $f(m)^2 = e$ is only satisfied if $f(m) = e$. Thus the only possible is the identity homomorphism.

2. Find all homomorphisms $f : D_n \rightarrow C_m$.

Solution: Write $D_n = \{e, r, r^2, \dots, r^{n-1}, m, rm, r^2m, \dots, r^{n-1}m\}$ and $C_m = \{e, s, s^2, \dots, s^{m-1}\}$. A homomorphism $f : D_n \rightarrow C_m$ is then determined entirely by the values of $f(r)$, $f(m)$, subject to them respecting the relations of D_n .

The relations we require are $f(r)^n = e$, $f(m)^2 = e$ and $f(r)f(m) = f(m)f(r)^{-1}$. In C_m , the third relation can be rewritten by commuting $f(m)$ past $f(r)$ to give $f(r) = f(r)^{-1}$, and hence $f(r)^2 = e$. Equivalently then, we require $f(r)^n = e$, $f(m)^2 = e$ and $f(r)^2 = e$.

We now divide into cases:

- If n, m are odd, the only way we can satisfy the relations is to require that $f(r) = e$ and $f(m) = e$. This gives the trivial homomorphism.

- If n is even and m is odd, then there are no elements of order two in C_m , so we still require $f(r) = e$ and $f(m) = e$. This gives the trivial homomorphism.
- If n is odd, and m is even, then we require $f(r) = e$ and can choose either $f(m) = e$ or $f(m) = s^{m/2}$. This gives rise to two homomorphisms.
- If n is even, and m is even, then we require $f(r) = e$ or $f(r) = s^{m/2}$, and either $f(m) = e$ or $f(m) = s^{m/2}$. This gives rise to four possible homomorphisms.

3. We say that a map $f : A \rightarrow B$ is *surjective* if for every element $b \in B$, there is some element $a \in A$ such that $f(a) = b$. Show that if $f : D_3 \rightarrow G$ is a surjective homomorphism, then G must be isomorphic to one of D_3, C_2 or C_1 .

Solution: The homomorphism is determined by $f(r), f(m)$ subject to the relations $f(r)^3 = e$, $f(m)^2 = e$ and $f(r)f(m) = f(m)f(r)^{-1}$. For the homomorphism to be surjective, we require $|G| \leq 6$ else we could not hit everything in G . Furthermore, we know that the image $\text{Im}(f)$ is a subgroup of the image, so by Lagrange's theorem must have size either 1, 2, 3 or 6. Checking each case in turn then, we have:

- In the case $|G| = 1$, the trivial homomorphism $f(r) = f(m) = e$ satisfies all relevant conditions and is the only possibility.
- In the case $|G| = 2$, we have $G = \{e, s\}$. We must have $f(r) = e$, and $f(m) = e$ or $f(m) = s$, giving two possible homomorphisms. The second possible homomorphism with $f(r) = e$ and $f(m) = s$ is clearly surjective.
- In the case $|G| = 3$, we have $G = \{e, s, s^2\}$. In this case, we require $f(m) = e$, and hence $f(r) = f(r)^{-1}$. But the only way for the second condition to hold is for $f(r) = e$. Thus the only possibility is the trivial homomorphism, which is not surjective.
- In the case $|G| = 6$, the groups in question are either $G \cong D_3$ or $G \cong C_6 = \{e, s, s^2, s^3, s^4, s^5\}$. There is an obvious surjective homomorphism $f : D_3 \rightarrow D_3$ given by just sending all elements to themselves. In the second case, we require $f(m) = e$ or $f(m) = s^3$. Since the group is Abelian in the second case, we have $f(r)f(m) = f(m)f(r)^{-1}$ implies that $f(r)^2 = e$, which gives $f(r) = e$ or $f(r) = s^3$. But then all possible products of $f(r), f(m)$ and their inverses can only give $\{e, s^3\}$, and hence the map is not surjective.

4. We say that a map $f : A \rightarrow B$ is *injective* if $f(a) = f(a')$ implies $a = a'$; that is, no two distinct elements in A map to the same element in B . Show that $f : G \rightarrow H$ is an injective homomorphism between the groups G, H if and only if the kernel $\text{Ker}(f)$ is trivial.

Solution: Suppose that $f : G \rightarrow H$ is injective, and let $k \in \text{Ker}(f)$. Then we must have $f(k) = e = f(e)$, since the identity of a group is mapped to the identity by a homomorphism. But now by injectivity, $k = e$, so that the kernel is trivial.

Conversely, suppose that $\text{Ker}(f) = \{e\}$. Then if $f(a) = f(a')$, we must have $f(a)f(a')^{-1} = e$. Using the preservation of inverses, and the homomorphism property, this gives $f(a(a')^{-1}) = e$, so that $a(a')^{-1}$ is in the kernel. Thus $a(a')^{-1} = e$, giving $a = a'$. So f is injective, as required.

5. Let $\text{Cl}(x) \subseteq G$ be the conjugacy class of an element $x \in G$ of a group G . Suppose that $f : G \rightarrow H$ is a homomorphism. Show that $f(\text{Cl}(x)) \subseteq \text{Cl}(f(x))$. Must this hold as an equality?

Solution: Let $y \in \text{Cl}(x)$. Then we have $y = gxg^{-1}$ for some $g \in G$, and hence $f(y) = f(gxg^{-1}) = f(g)f(x)f(g)^{-1}$. Thus $f(y)$ is a conjugate of $f(x)$, and it follows that $f(y) \in \text{Cl}(f(x))$. Thus indeed $f(\text{Cl}(x)) \subseteq \text{Cl}(f(x))$.

This need not hold as an equality; consider for example $f : C_2 \rightarrow D_3$, where $C_2 = \{e, s\}$, $D_3 = \{e, r, r^2, m, rm, r^2m\}$, given by $f(e) = e$ and $f(s) = m$. This is certainly a homomorphism since the relation $f(s)^2 = m^2 = e$ on s is preserved. The conjugacy classes in C_2 are $\{e\}, \{s\}$. Meanwhile the conjugacy classes in D_3 are $\{e\}, \{r, r^2\}, \{m, rm, r^2m\}$. Hence $\{f(s)\} = \{m\} \subseteq \{m, rm, r^2m\}$ is strictly contained in the image conjugacy class.

The isomorphism theorem

6. Define the sets of matrices:

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a, b \in \mathbb{C}, a \neq 0 \right\}, \quad H = \left\{ \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} : a \in \mathbb{C}, a \neq 0 \right\}.$$

(a) Show that G is a group, and show that $H \leq G$ is a subgroup of G .

(b) Prove that the map $f : G \rightarrow H$ given by:

$$f \left(\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

is a homomorphism, and hence identify a non-trivial normal subgroup K of G .

(c) Show that $G/K \cong \mathbb{C}^*$.

Solution:

(a) For the first part, we just verify the group axioms in each case. For G , we have:

(G1) *Closure.* Given two elements of G , their product takes the form:

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} aa' & ab' + b \\ 0 & 1 \end{pmatrix},$$

which is in G , since $a \neq 0, a' \neq 0$ implies that $aa' \neq 0$.

(G2) *Associativity.* Matrix multiplication is associative.

(G3) *Identity.* The identity is contained in G , with $a = 1, b = 0$.

(G4) *Inverses.* The inverse of a given element is:

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{a} \begin{pmatrix} 1 & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/a \\ 0 & 1 \end{pmatrix},$$

which is still contained in G .

Therefore G is a group. Looking at the above axioms and taking $b = b' = 0$, we see that H also forms a group.

(b) The given map is a homomorphism since:

$$\begin{aligned}
 f\left(\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & 1 \end{pmatrix}\right) &= f\left(\begin{pmatrix} aa' & ab' + b \\ 0 & 1 \end{pmatrix}\right) \\
 &= \begin{pmatrix} aa' & 0 \\ 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a' & 0 \\ 0 & 1 \end{pmatrix} \\
 &= f\left(\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\right) f\left(\begin{pmatrix} a' & b' \\ 0 & 1 \end{pmatrix}\right).
 \end{aligned}$$

A non-trivial normal subgroup is given by the kernel of this homomorphism, which is the set:

$$K = \text{Ker}(f) = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{C} \right\}.$$

(c) The isomorphism theorem tells us that $G/K \cong \text{Im}(f)$. The image is the set of matrices of the form:

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix},$$

where a is a non-zero complex number. This set of matrices is clearly isomorphic to $\mathbb{C}^*$.

7. By identifying an appropriate normal subgroup $K \trianglelefteq T$, find a non-trivial homomorphism $f : T \rightarrow C_3$, where T is the chiral tetrahedral group.

Solution: The only proper non-trivial normal subgroup is the subgroup $K = \{I, R, G, B\}$ consisting of the identity and the rotations by 180° through axes that pass through the midpoints of opposite edges. Since $T/K \cong C_3$ (consider sizes), by the isomorphism theorem there is an isomorphism $g : T/K \rightarrow C_3$. We lift this to a homomorphism $f : T \rightarrow C_3$ by setting:

$$f(x) = g(xK).$$

This is a homomorphism since $f(xy) = g((xy)K) = g(xKyK) = g(xK)g(yK) = f(x)f(y)$. The kernel of f is evidently K , and we have already shown that $T/K \cong C_3$.

8. Let $SL(n, \mathbb{R})$ be the set of all invertible $n \times n$ matrices whose determinant is equal to one. By finding an appropriate homomorphism $f : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$, show that $SL(n, \mathbb{R}) \leq GL(n, \mathbb{R})$ is a normal subgroup of $GL(n, \mathbb{R})$, and that $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^*$.

Solution: Use the homomorphism $\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$. This is a homomorphism since $\det(AB) = \det(A)\det(B)$ for all matrices A, B . The kernel of the homomorphism is the preimage of $1 \in \mathbb{R}^*$, which is precisely the subset of $GL(n, \mathbb{R})$ of matrices with determinant 1, namely $SL(n, \mathbb{R})$. Since $SL(n, \mathbb{R})$ is the kernel of a homomorphism, it is a normal subgroup of $GL(n, \mathbb{R})$.

Note that the homomorphism $\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is surjective since we can have matrices of all possible determinants. Thus $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^*$ by the isomorphism theorem.

Automorphisms

9. Identify standard groups to which $\text{Aut}(D_2)$ and $\text{Aut}(D_3)$ are isomorphic.

Solution: We consider both cases separately:

- Let $D_2 = \{e, r, m, rm\}$. An automorphism $\alpha : D_2 \rightarrow D_2$ is determined by $\alpha(r), \alpha(m)$. Since both $\alpha(r), \alpha(m)$ need to have order 2, they must be mapped to two distinct elements drawn from r, m or rm . This gives a total of 6 choices.

We shall show that any such choice produces an automorphism. Certainly it is true that if $\alpha(r), \alpha(m)$ are chosen to be distinct elements from r, m, rm , then their product is the third; this is how multiplication works in D_2 . Therefore the map is always bijective. Furthermore, the generators satisfy the relations $r^2 = e, m^2 = e$ and $rm = mr$. Since the group is Abelian, the third is satisfied automatically by $\alpha(r), \alpha(m)$. Further the first two are satisfied automatically by the choice of them being elements drawn from r, m, rm . It follows that any such choice indeed produces an automorphism.

Finally, we can imagine that r, m, rm are the vertices of a triangle. Any automorphism permutes these vertices, so can be put into one-to-one-correspondence with the elements of D_3 . In particular, this shows that $\text{Aut}(D_2) \cong D_3$.

- Let $D_3 = \{e, r, r^2, m, rm, r^2m\}$. An automorphism $\alpha : D_3 \rightarrow D_3$ is determined by $\alpha(r), \alpha(m)$. Since $\alpha(r)$ needs to have order 3, it must be either r or r^2 . Since $\alpha(m)$ needs to have order 2, it must be one of m, rm, r^2m . Thus there are potentially six possibilities.

Since D_3 is generated by any of the rotations, and any of the reflections, we have mapped $\{r, m\}$ to a generating set $\{\alpha(r), \alpha(m)\}$. Consequently, we can write any element in the form $\alpha(r)^a \alpha(m)^b = \alpha(r^a m^b)$ if this is a homomorphism, and hence this is a surjection.

To check choosing $\alpha(r) \in \{r, r^2\}$ and $\alpha(m) \in \{m, rm, r^2m\}$ gives a homomorphism, we note that the relations $\alpha(r)^3 = e$ and $\alpha(m)^2 = e$ are immediate. Further $\alpha(r)\alpha(m) = \alpha(m)\alpha(r)^{-1}$ holds from the standard relation between rotations and reflections. Thus this is indeed a homomorphism, and hence all six possibilities result in an automorphism.

The automorphism group is therefore isomorphic to a group of order six, either C_6 or D_3 . Consider the automorphisms α, β generated by:

$$\alpha(r) = r, \quad \alpha(m) = rm, \quad \text{and} \quad \beta(r) = r^2, \quad \beta(m) = m.$$

Observe that:

$$\alpha(\beta(m)) = \alpha(m) = rm, \quad \beta(\alpha(m)) = \beta(rm) = \beta(r)\beta(m) = r^2m.$$

Hence α, β do not commute, so the group structure is non-Abelian. It follows that the automorphism group is D_3 . In particular, this shows that $\text{Aut}(D_3) \cong D_3$ too.

10. Show that $\text{Aut}(C_n) \cong \mathbb{Z}_n^*$.

Solution: Suppose that $\alpha : C_n \rightarrow C_n$ is an automorphism. Since r has order n , we must map to some element $r^k \in C_n$ with order dividing n . But all elements have this property in C_n . Hence $\alpha_k(r) = r^k$ is a potential automorphism for each k ; checking the homomorphism property, we have:

$$\alpha_k(r^a r^b) = \alpha_k(r^{a+b}) = r^{k(a+b)} = r^{ka} r^{kb} = \alpha_k(r^a) \alpha_k(r^b).$$

To ensure this is a surjection, we must be able to generate r from powers of r^k ; if we can do so, we can then generate the entire group. That is, there must exist some m such that $r^{km} = r$, i.e. $km = 1$ modulo n . This requires that k is invertible modulo n . Thus the automorphisms precisely take the form:

$$\alpha_k(r^a) = r^{ka},$$

where $k \in \mathbb{Z}_n^*$.

To finish showing that the automorphism group is isomorphic to $\mathbb{Z}_n^*$, we define an isomorphism $f : \text{Aut}(C_n) \rightarrow \mathbb{Z}_n^*$ by $f(\alpha_k) = k$. We claim that this satisfies the two properties required of an automorphism:

(i) It is obviously invertible, by construction.

(ii) Note that:

$$\alpha_k \circ \alpha_l(r^a) = r^{kla} = \alpha_{kl}(r^a).$$

$$\text{Hence } f(\alpha_k \circ \alpha_l) = f(\alpha_{kl}) = kl = f(\alpha_k)f(\alpha_l).$$

Thus we conclude that $\text{Aut}(C_n) \cong \mathbb{Z}_n^*$, as required.

11. The set of all *inner automorphisms* $\text{Inn}(G)$ of a group G is the set of all automorphisms $\alpha_g : G \rightarrow G$ of the form:

$$\alpha_g(x) = gxg^{-1};$$

that is, it is the set of all automorphisms given by conjugation.

(a) Show that $\text{Inn}(G)$ is a normal subgroup of $\text{Aut}(G)$.

Solution: First, we show that $\text{Inn}(G)$ is a subgroup:

(G1) *Closure.* If α_g, α_h are both inner automorphisms, then:

$$(\alpha_g \circ \alpha_h)(x) = ghxh^{-1}g^{-1} = (gh)x(gh)^{-1} = \alpha_{gh}(x),$$

so their composition is an inner automorphism.

(G2) *Associativity.* Function composition is associative.

(G3) *Identity.* The identity is α_e , given by $\alpha_e(x) = exe^{-1} = x$.

(G4) *Inverses.* The inverse of α_g is $\alpha_{g^{-1}}$, since $\alpha_g \circ \alpha_{g^{-1}} = \alpha_{gg^{-1}} = \alpha_e$.

Further, this subgroup is normal. Suppose that β is any automorphism. Then we have:

$$(\beta \circ \alpha_g \circ \beta^{-1})(x) = \beta(g\beta^{-1}(x)\beta^{-1}) = \beta(g)x\beta(g^{-1}) = \beta(g)x\beta(g)^{-1} = \alpha_{\beta(g)}(x).$$

Thus $\beta \circ \alpha_g \circ \beta^{-1} = \alpha_{\beta(g)} \in \text{Inn}(G)$. It follows that indeed $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

(b) Show that:

$$\text{Inn}(G) \cong \frac{G}{Z(G)},$$

where $Z(G)$ is the centre of the group G , defined in Examples Sheet 5, Question 4.

Solution: Define $f : G \rightarrow \text{Inn}(G)$ given by $f(g) = \alpha_g$. Then certainly this map is surjective, and it is also a homomorphism:

$$f(gh) = \alpha_{gh} = \alpha_g \circ \alpha_h = f(g)f(h).$$

Thus by the isomorphism theorem, we have $G/\text{Ker}(f) \cong \text{Inn}(G)$. It remains to show that $\text{Ker}(f) = Z(G)$.

Suppose that $k \in \text{Ker}(f)$. Then we have $f(k) = \alpha_k = \text{id}_G$ (the identity automorphism), hence:

$$kxk^{-1} = x$$

for all $x \in G$, and thus $k \in Z(G)$. Conversely, if $k \in Z(G)$, we have $kxk^{-1} = x$ for all $x \in G$, so $\alpha_k = \text{id}_G$. Thus the kernel is exactly $Z(G)$, as required.

(c) Identify standard groups to which $\text{Inn}(C_n)$, $\text{Inn}(D_2)$ and $\text{Inn}(D_3)$ are isomorphic.

Solution: Using the above result:

- (a) The centre of C_n is C_n itself, since C_n is Abelian. Thus $C_n/C_n \cong C_1$, thus $\text{Inn}(C_n) \cong C_1$.
- (b) Similarly for D_2 , since D_2 is Abelian, we must have $\text{Inn}(D_2) \cong C_1$.
- (c) The centre of D_3 is just $\{e\}$, so $\text{Inn}(D_3) \cong D_3$.

The set of all *outer automorphisms* is defined to be $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$.

(d) Identify standard groups to which $\text{Out}(C_n)$, $\text{Out}(D_2)$ and $\text{Out}(D_3)$ are isomorphic.

Solution: We have $\text{Out}(C_n) \cong \mathbb{Z}_n^*/C_1 \cong \mathbb{Z}_n^*$. Similarly we have $\text{Out}(D_2) \cong D_3/C_1 \cong D_3$. Finally, we have $\text{Out}(D_3) \cong D_3/D_3 \cong C_1$.

12. Show that $\text{Aut}(Q_8) \cong O$, where O is the chiral cubic group. Identify standard groups to which $\text{Inn}(Q_8)$ and $\text{Out}(Q_8)$ are isomorphic.

Solution: Let $\alpha : Q_8 \rightarrow Q_8$ be an automorphism. Certainly α must fix $1, -1$, since the identity must be fixed by an automorphism, and -1 must be fixed as it is the unique element of order two in Q_8 .

Note that the group Q_8 is generated by i, j subject to the relations $i^2 = j^2, i^4 = 1$, and $jij^{-1} = i^{-1}$. By the above argument, the automorphism α must send i, j to two distinct elements of $S = \{\pm i, \pm j, \pm k\}$. If i, j are sent to $i, -i$ say, then we would have $f(j)f(i)f(j)^{-1} = (-i)i(i) = i \neq -i = f(i)^{-1}$, so we would contradict one of the relations. Similar conclusions happen if we choose i, j to map to $-i, i$ respectively, or any other pairs $\{j, -j\}, \{k, -k\}$.

Thus, we have six choices for i , and once we have picked its value, we must choose j from one of the two pairs that are not related by a $\pm$ sign to the image of i . This gives a total of 24 possible choices for where to send i, j . This is important because O has order 24, so this must be all of the automorphisms. Checking that indeed these maps f are automorphisms:

- Certainly $f(i)^2 = f(j)^2$ since all elements of the set S square to -1 . Further $f(i)^4 = 1$, since all elements of the set S have fourth power 1. The final relation is:

$$f(j)f(i)f(j)^{-1} = f(i)^{-1}.$$

The inverse of any element in S is its negative, which implies we need $f(j)f(i)f(j) = f(i)$ to hold. Multiplying by $f(j)^{-1} = -f(j)$ on the right, equivalently we need $f(j)f(i) = -f(i)f(j)$. This holds for any two elements of S , provided that they are not a pair related by a sign, e.g. $\{\pm i\}$. Hence all conditions for f being a homomorphism are satisfied.

- Next, observe that f is a surjection. This is because any two of S , not a pair related by a sign, generate Q_8 .

To prove the automorphism group is isomorphic to O , observe that we may imagine $\pm i, \pm j, \pm k$ as the faces of a cube. Any automorphism corresponds to some permutation of the faces. First, we must choose where i can go - this corresponds to moving a particular face. We are not free to choose $-i$ though; this means the face $-i$ is fixed opposite to i . Finally, we are free to choose j to go anywhere of four places, corresponding to the adjacent faces to i . This gives the structure of O .

For $\text{Inn}(Q_8) \cong Q_8/Z(Q_8)$, we observe that $Z(Q_8) = \{\pm 1\}$, hence $Q_8/Z(Q_8) = \{Z(Q_8), iZ(Q_8), jZ(Q_8), kZ(Q_8)\} \cong D_2$. For the outer automorphism group, we first need to identify D_2 inside O . Note that O has three elements of order two, namely the rotations by 180° around each of the axes through the faces. These are all commuting, and isomorphic to D_2 .

Quotienting by 180° rotations effectively identifies opposite faces of the cube together. But this gives three pairs of opposite faces, which can be arbitrarily permuted. Thus the quotient must be:

$$\text{Out}(Q_8) \cong D_3.$$